



YOUR FIRST MACHINE LEARNING MODELS

HOW TO BUILD THEM IN SCIKIT-LEARN



with Corey Wade, Director & Founder of Berkeley Coding Academy

OVERVIEW

- ▶ Prerequisites - Comfortable Coding in Python
- ▶ Requirements - Jupyter Notebook
 - ▶ Recommend - <https://www.anaconda.com/download>
 - ▶ Contains required Data Science libraries - pandas, sklearn, numpy
- ▶ Clone Repository - github.com/coreyjwade/pycon2026
- ▶ Open Jupyter Notebooks and open "Starter_Your_First_ML_Models"

WORKSHOP OUTLINE

- Overview of Machine Learning landscape
- Build and score Linear Regression model
- Classify data with Logistic Regression
- Overfitting and adjusting parameters with Decision Trees
- Splitting data with cross-validation and changing scoring metrics
- Ensembling data with Random Forests and Gradient Boosting
- Fine-tuning models with GridSearchCV and RandomizedSearchCV
- Finalizing and saving models for future reference

WORKSHOP DOES NOT INCLUDE

- ▶ Unstructured datasets (with images and text)
 - ▶ Generative AI
 - ▶ Large Language Models
- ▶ Neural networks (great next step)
- ▶ Unsupervised Learning
 - ▶ This is a workshop in Supervised Learning where models are trained with known outcomes



MEET YOUR INSTRUCTOR

Corey Wade

- ▶ Director & Founder of Berkeley Coding Academy: <https://www.berkeleycodingacademy.com/>
 - ▶ Teach data science to teenagers with a focus on machine learning
- ▶ Author of *Hands-on Gradient Boosting with XGBoost and Scikit-learn, The Python Workshop*.
- ▶ Math/Data Science teacher at Berkeley Independent Study (BUSI).
- ▶ Starting in summer of 2026 as a Live Instructor with O'Reilly

WHAT IS MACHINE LEARNING?



MYSTERY FUNCTION 1

Input

Output

1

3

2

6

3

9

4

12

5

?



MYSTERY FUNCTION 1

Input

Output

1

3

2

6

3

9

4

12

5

15



MYSTERY FUNCTION 1

Triple the input.

$$f(x) = 3x$$

$$f = 3x$$

Predict new values:

$$f(6) = 3 \cdot 6$$

$$f(6) = 18$$

x	y
1	3
2	6
3	9
4	12
5	15

MYSTERY FUNCTION 2

Input	Output
1	2.65
2	6.23
3	9
4	11.814
5	?



MYSTERY FUNCTION 2

Input

Output

1

2.65

2

6.23

3

9

4

11.814

5

15



MYSTERY FUNCTION 2

Triple the input.

$$f(x) \approx 3x$$

$$f \approx 3x$$

Predict new values:

$$f(-2) \approx 3 \cdot (-2)$$

$$f(-2) \approx -6$$

x	y
1	2.65
2	6.23
3	9
4	11.814
5	15



MYSTERY FUNCTION 3

X1	X2	y
1	-1	3
2	1	2.95
3	-1	7.03
4	0	8
5	1	?



MYSTERY FUNCTION 3

X1	X2	y
1	-1	3
2	1	2.95
3	-1	7.03
4	0	8
5	1	9



MYSTERY FUNCTION 3

Double X1 then add the negative of X2.

$$f(x_1, x_2) \approx 2x_1 - 1x_2$$

$$f \approx 2x_1 - 1x_2$$

Predict new values:

$$f(6, -1) = 2 \cdot 6 - 1 \cdot (-1)$$

$$f(6) = 13$$

X1	X2	y
1	-1	3
2	1	2.95
3	-1	7.03
4	0	8
5	1	9



MYSTERY FUNCTION 4

total_bedrooms	population	households	median_income	median_house_value
1283.0	1015.0	472.0	1.4936	66900.0
1901.0	1129.0	463.0	1.8200	80100.0
174.0	333.0	117.0	1.6509	85700.0
337.0	515.0	226.0	3.1917	73400.0
326.0	624.0	262.0	1.9250	65500.0

We need a computer program that can find the coefficients for us.

Is this possible? Yes, it's called machine learning.

WHAT IS MACHINE LEARNING?

- ▶ Machine Learning takes functional data as input - **the X-values, and the y-value** -, and returns a mathematical model as output.
- ▶ The Machine Learning model fits itself to the data.
- ▶ It's called Machine Learning because the algorithm learns parameters from each row of data, and from new data coming in.
- ▶ Machine Learning is valuable because it makes predictions about the future when **X is known, and y is unknown**.
- ▶ There are many different types of Machine Learning algorithms. We will start with Linear Regression.

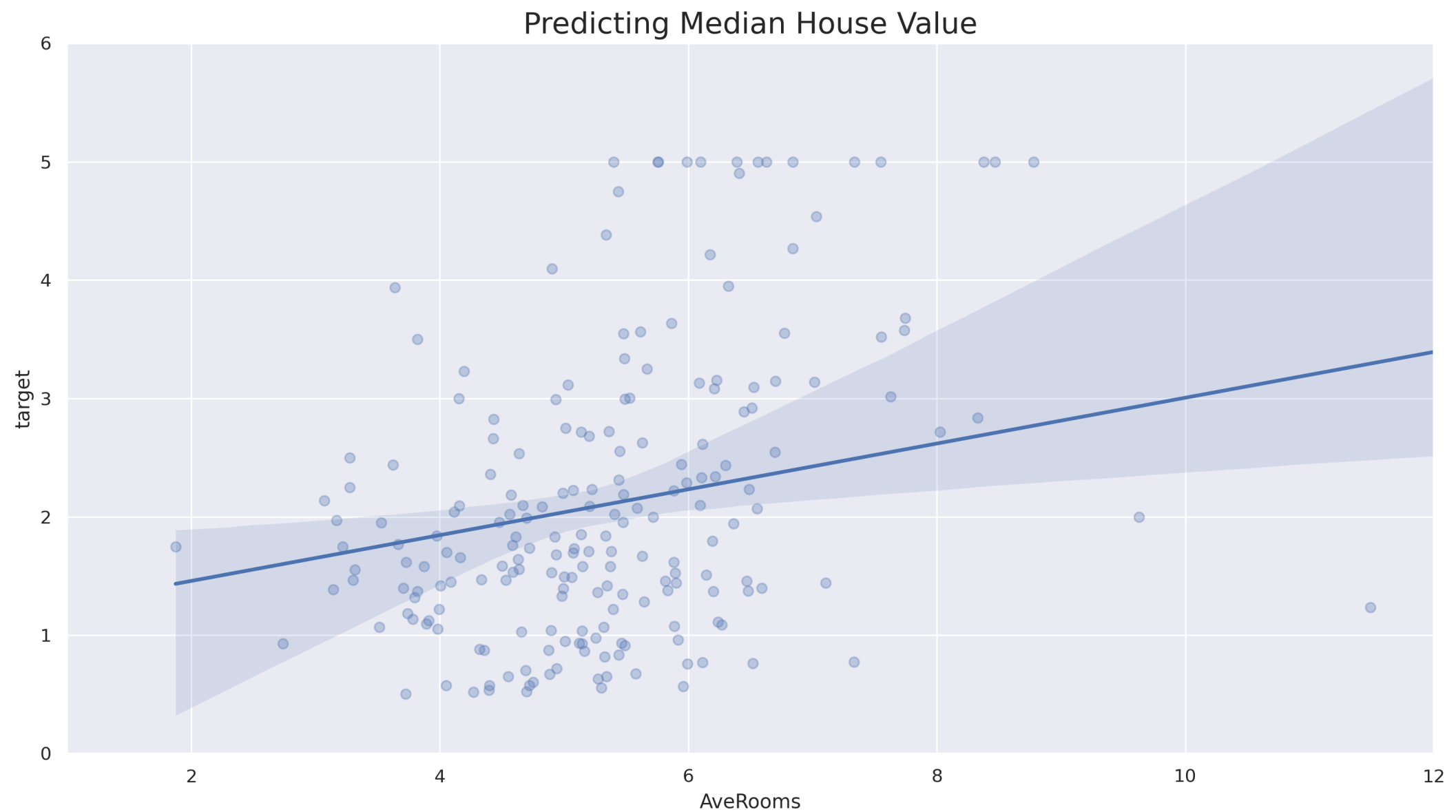
LINEAR REGRESSION

LINEAR REGRESSION

Given a set of data points, Linear Regression gives a straight line that minimizes the square of the distance between the points and the line.

Traditionally the least squares regression line was found using calculus.

LINEAR REGRESSION EXAMPLE





LINEAR REGRESSION WITH DATA

B_1 X_1 total_bedrooms	B_2 X_2 population	B_3 X_3 households	B_4 X_4 median_income	B_0 y median_house_value
1283.0	1015.0	472.0	1.4936	66900.0
1901.0	1129.0	463.0	1.8200	80100.0
174.0	333.0	117.0	1.6509	85700.0
337.0	515.0	226.0	3.1917	73400.0
326.0	624.0	262.0	1.9250	65500.0

You multiple each weight, B , by each col X , and sum to get as close as possible to y .

The key is to find the right B 's!



CHOOSING THE WEIGHTS

$$y = Bx$$

The weights, also known as the coefficients, are the values for B .

At first try, B is completely random.

After comparing the random predictions to the actual results, B is adjusted depending on whether the error is too high or too low using gradient descent (calculus).

A second iteration proceeds with the new values of B .

During each subsequent iteration, B is adjusted via gradient descent until the error is minimized.

After the error has been minimized for B , the final model is returned.

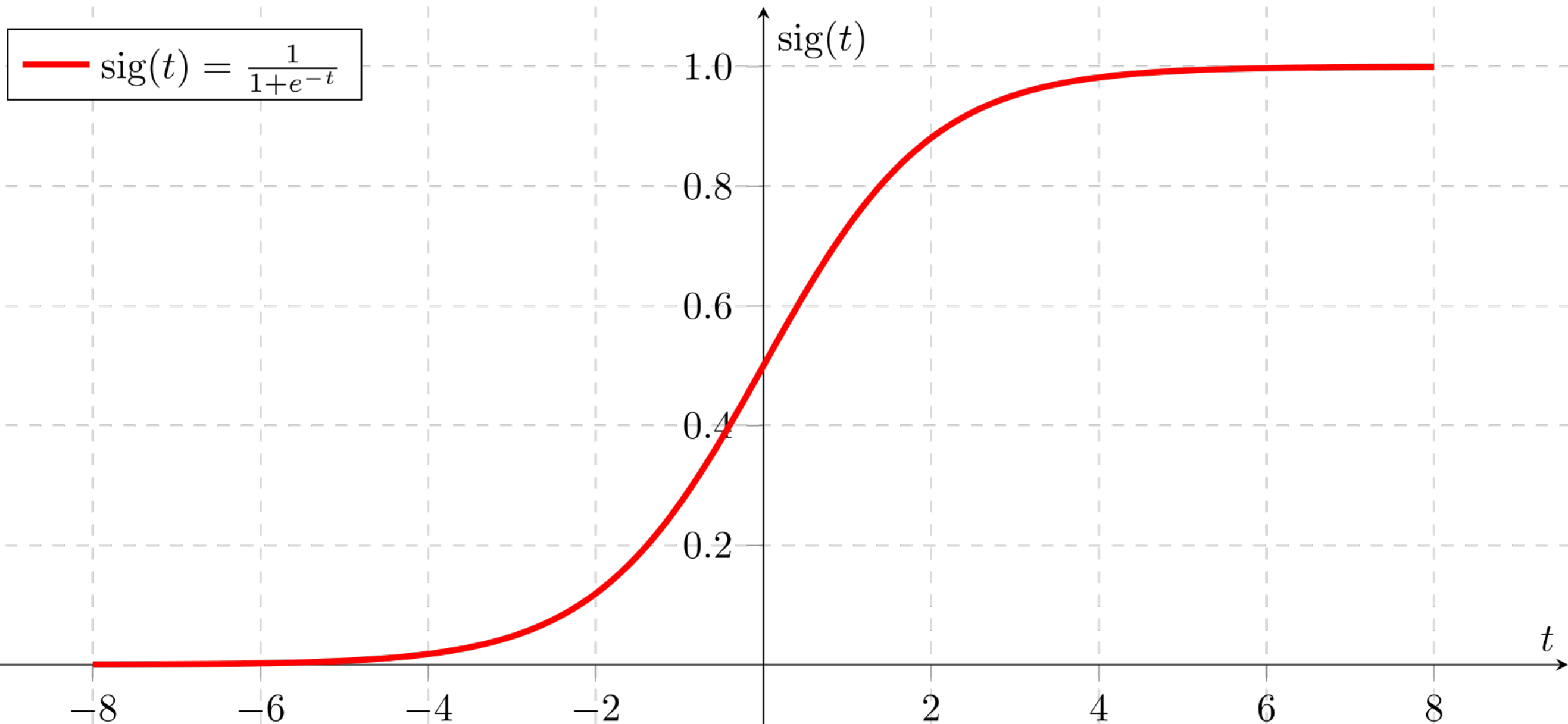
LOGISTIC REGRESSION

WHAT IS LOGISTIC REGRESSION?

- ▶ Logistic Regression is a standard classification machine learning algorithm.
- ▶ Like Linear Regression, Logistic Regression finds weights for each column of data.
- ▶ Unlike Linear Regression, Logistic Regression splits the data in half.
- ▶ Logistic Regression uses the sigmoid function to convert all outputs to 0 or 1.
- ▶ Logistic Regression is a classifier, not a regressor, even though it's called "Logistic Regression."



SIGMOID EQUATION

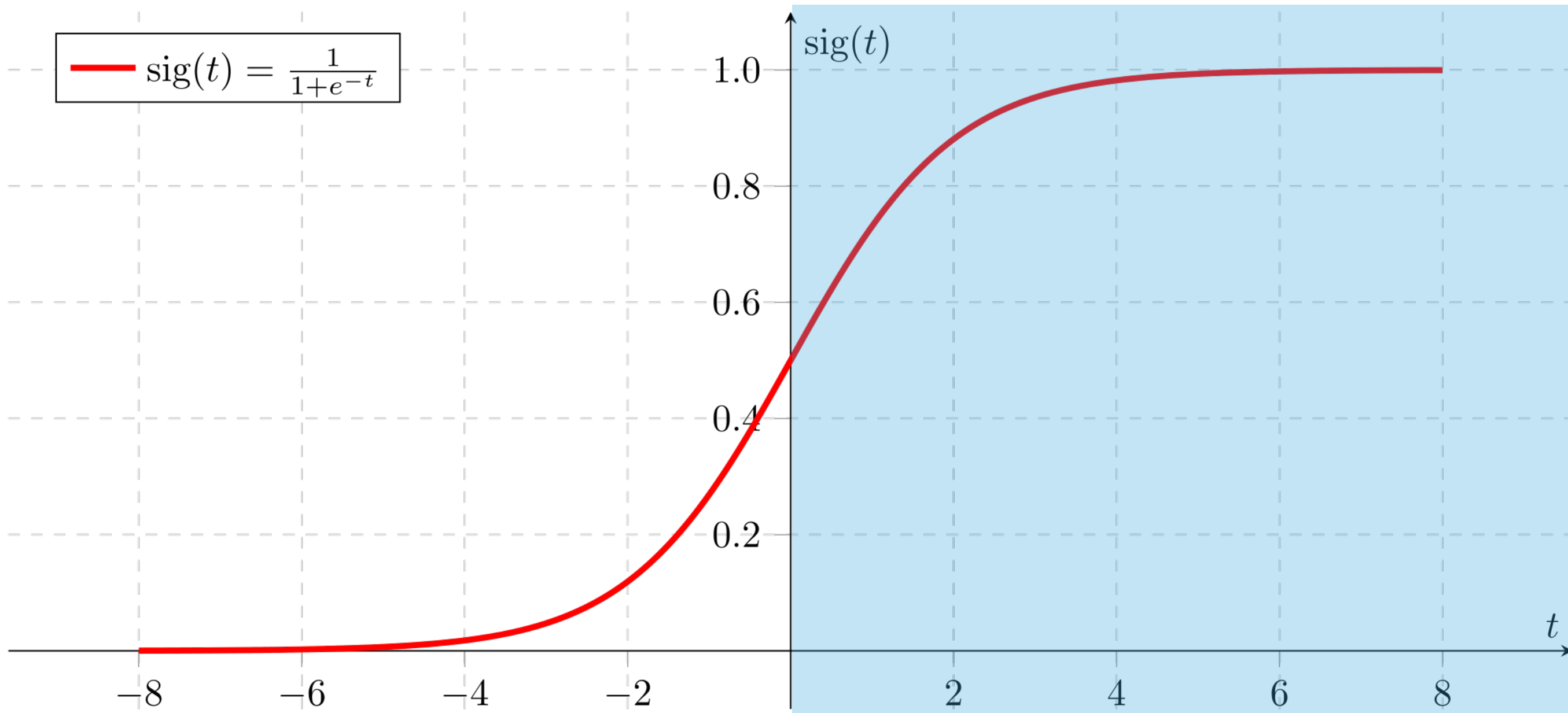




SIGMOID EQUATION

$$\text{sig}(t) = \frac{1}{1+e^{-t}}$$

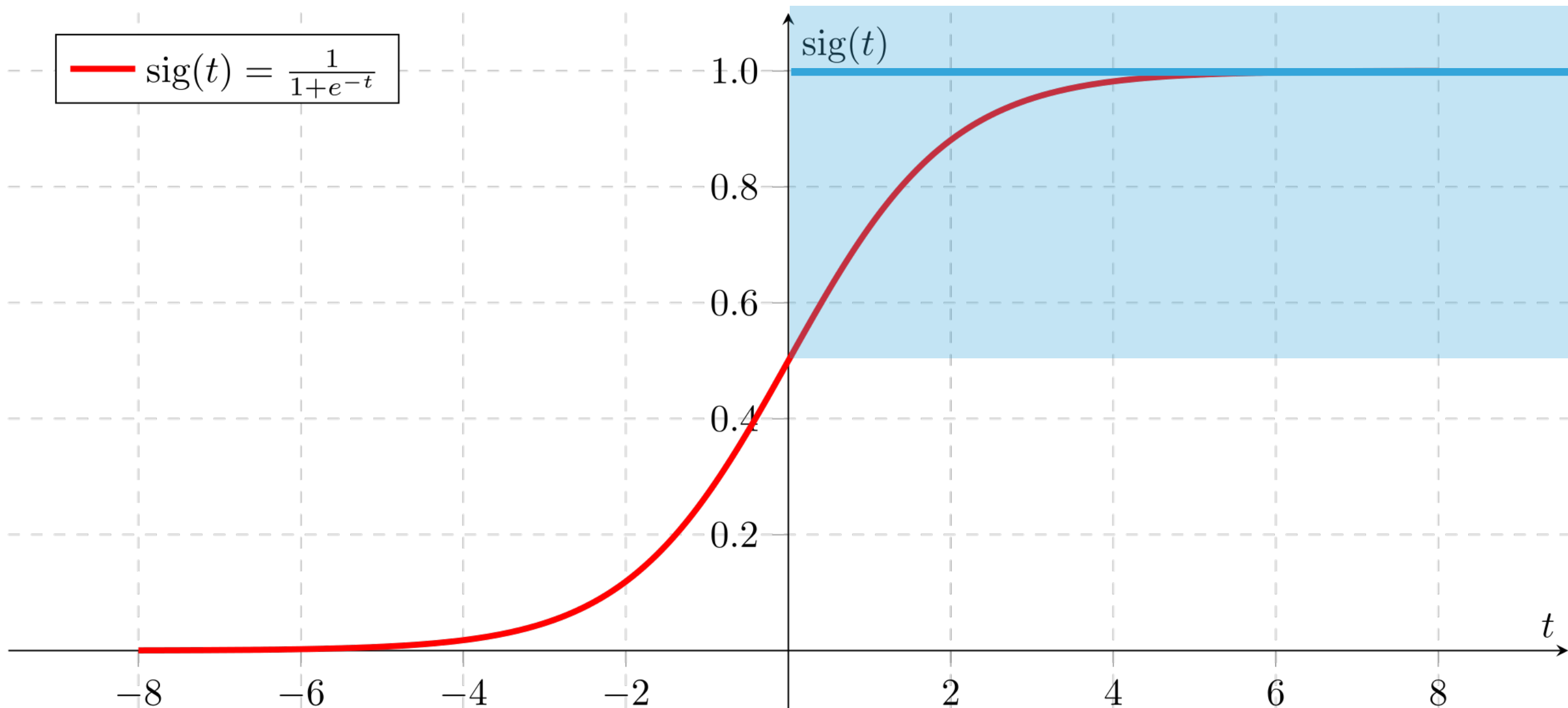
Positive X values



SIGMOID EQUATION

$$\text{sig}(t) = \frac{1}{1+e^{-t}}$$

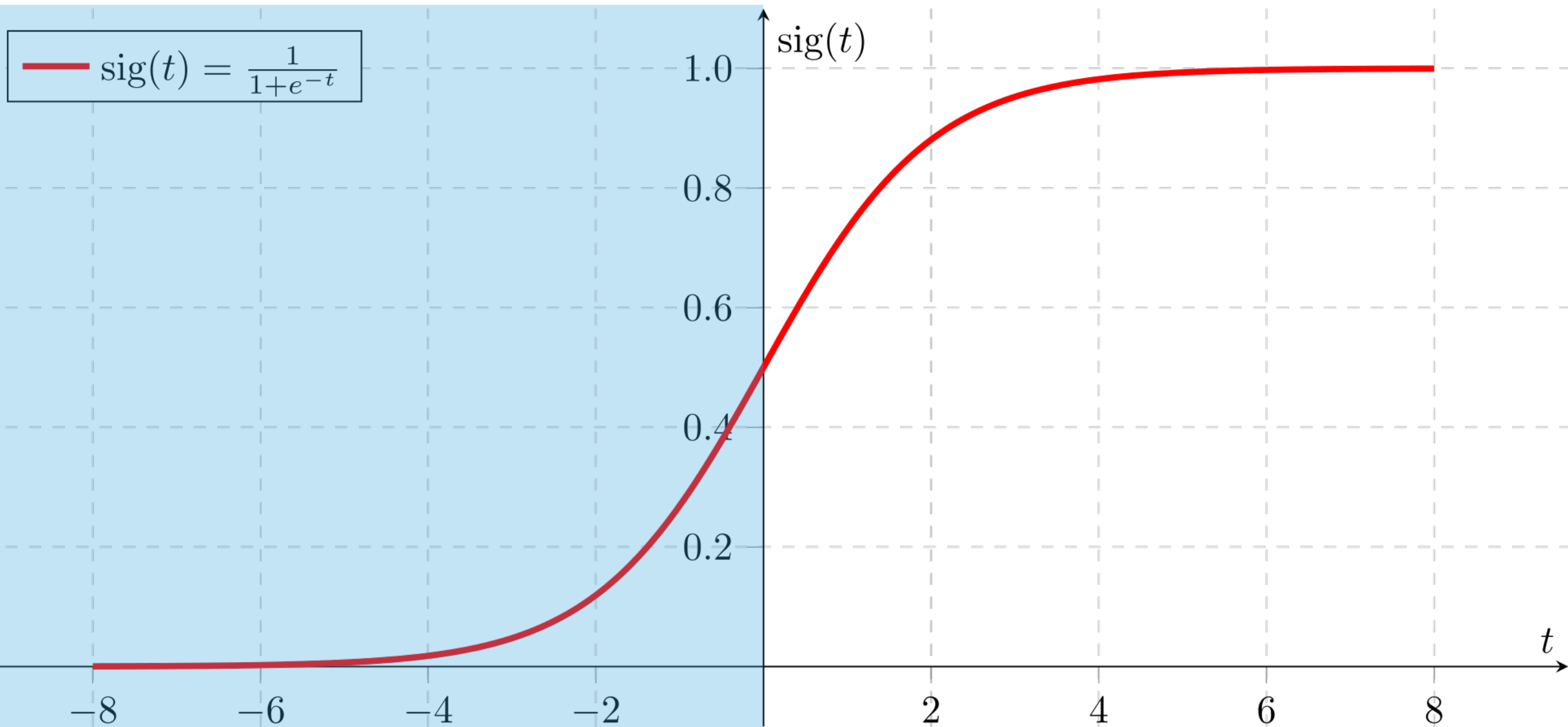
Positive X values are mapped to class 1





SIGMOID EQUATION

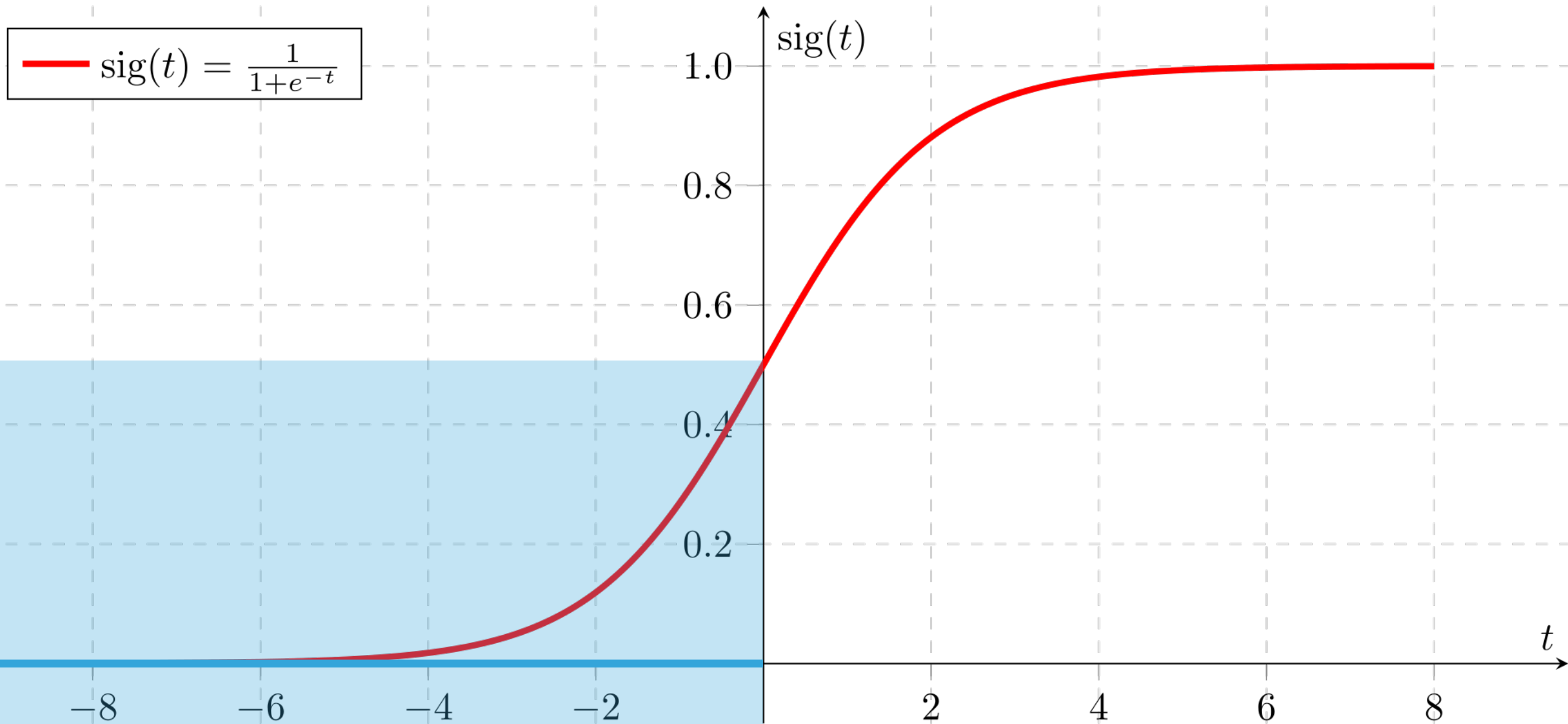
Negative X values





SIGMOID EQUATION

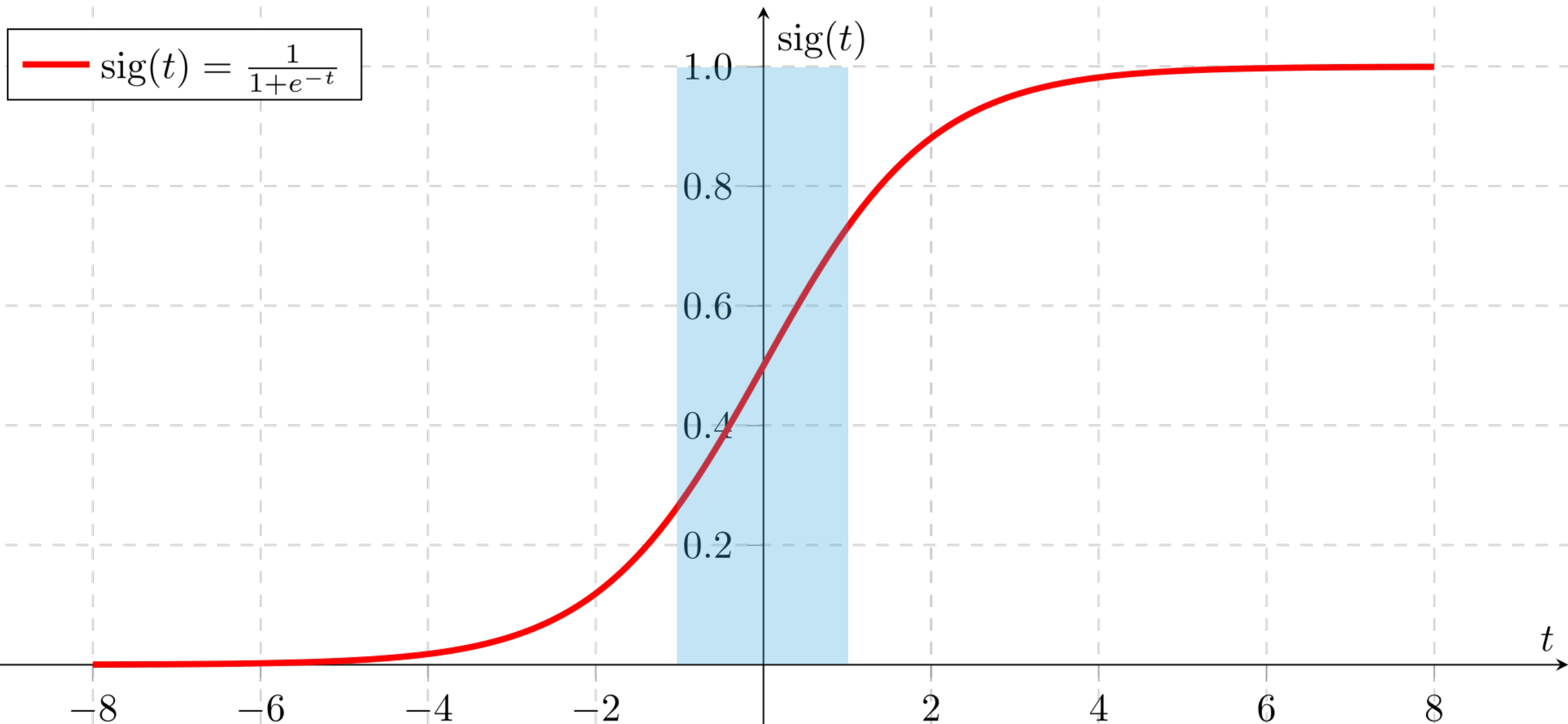
Negative X values are mapped to class 0





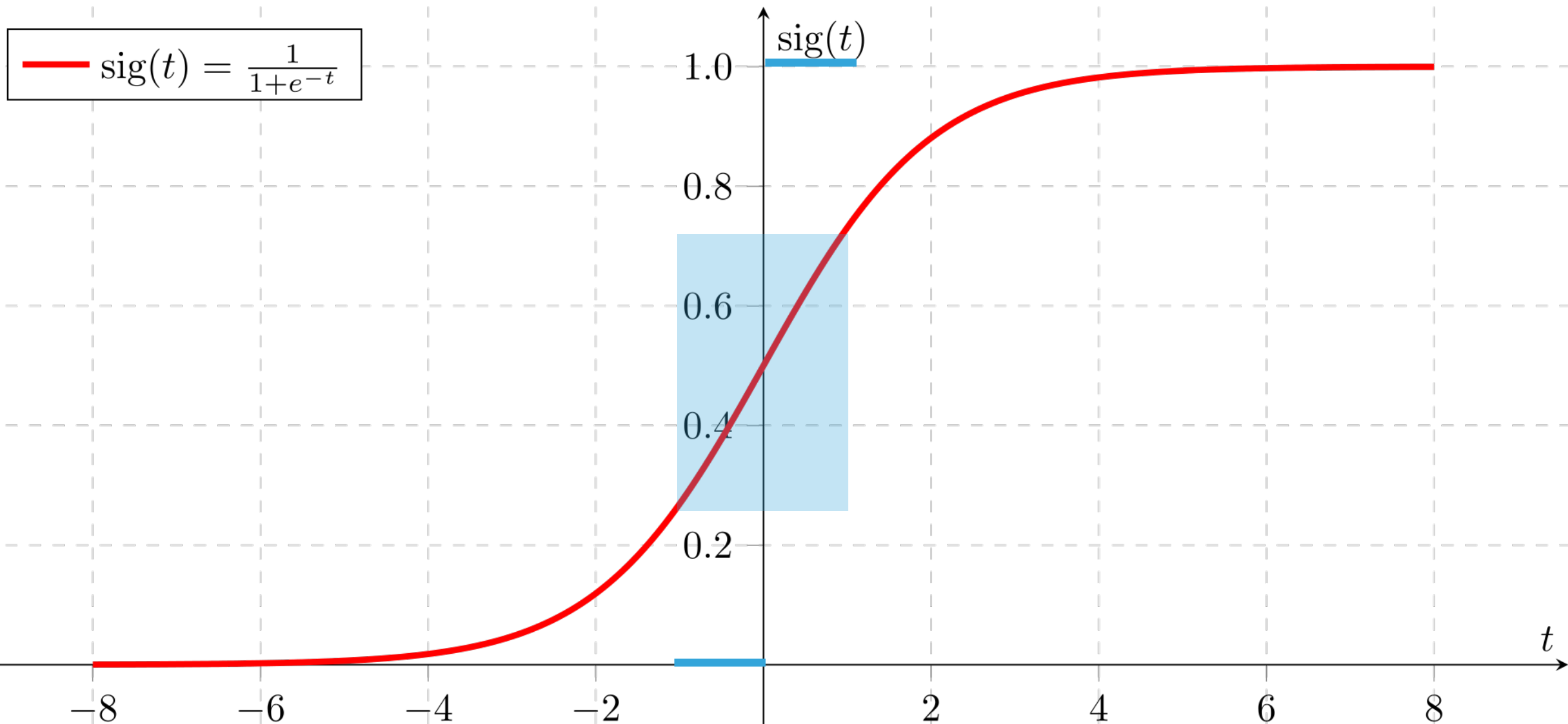
SIGMOID EQUATION

X values in the middle



SIGMOID EQUATION

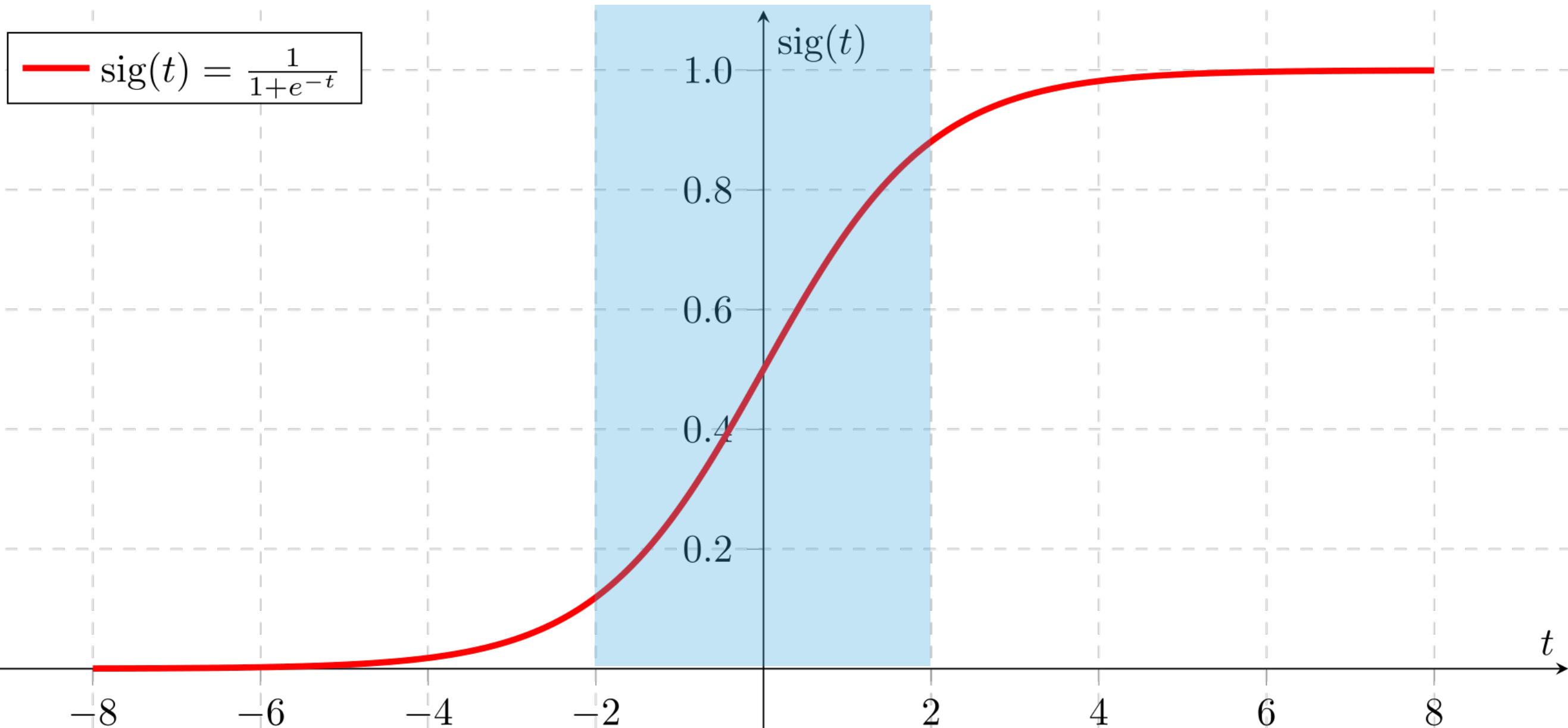
X values in the middle have the greatest uncertainty





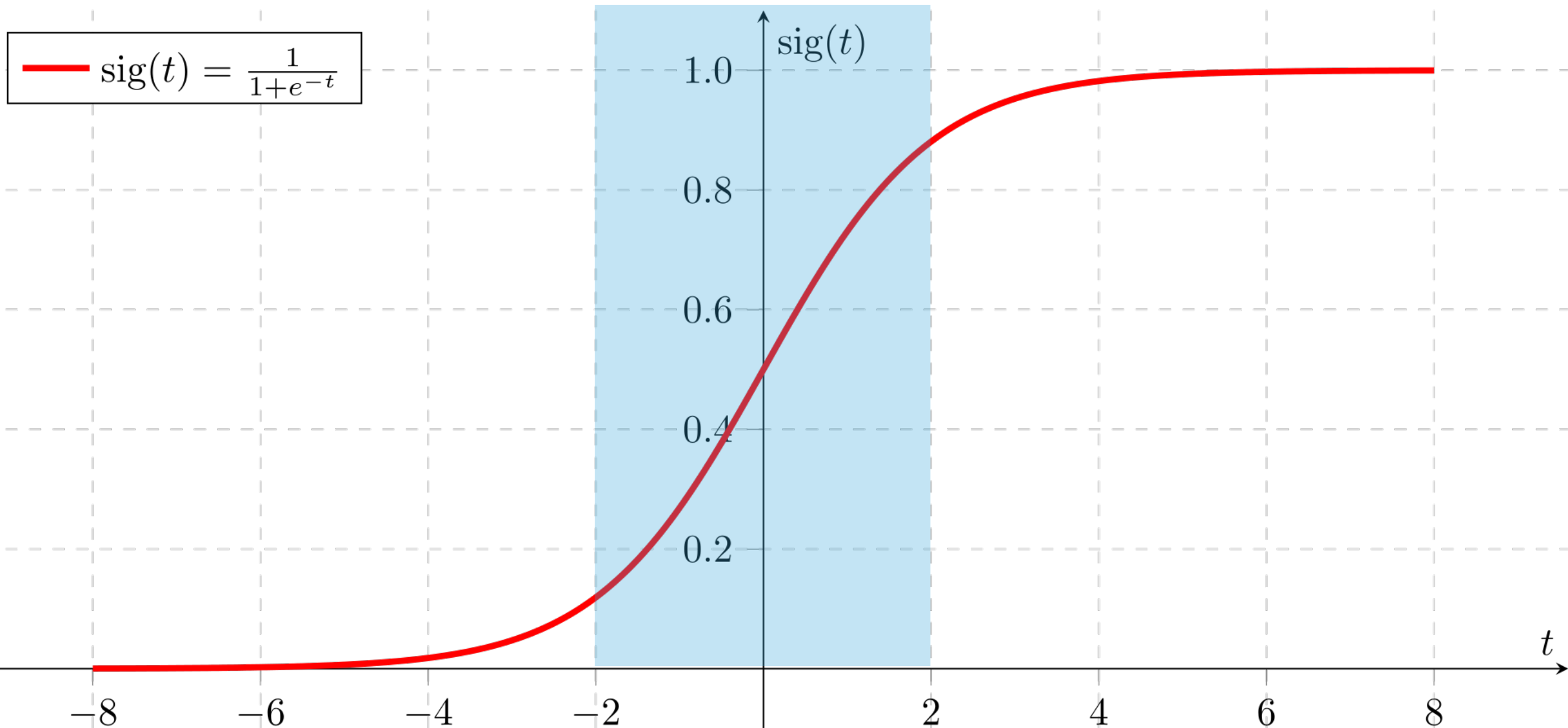
SIGMOID EQUATION

The Sigmoid ranges from 0 to 1



SIGMOID EQUATION

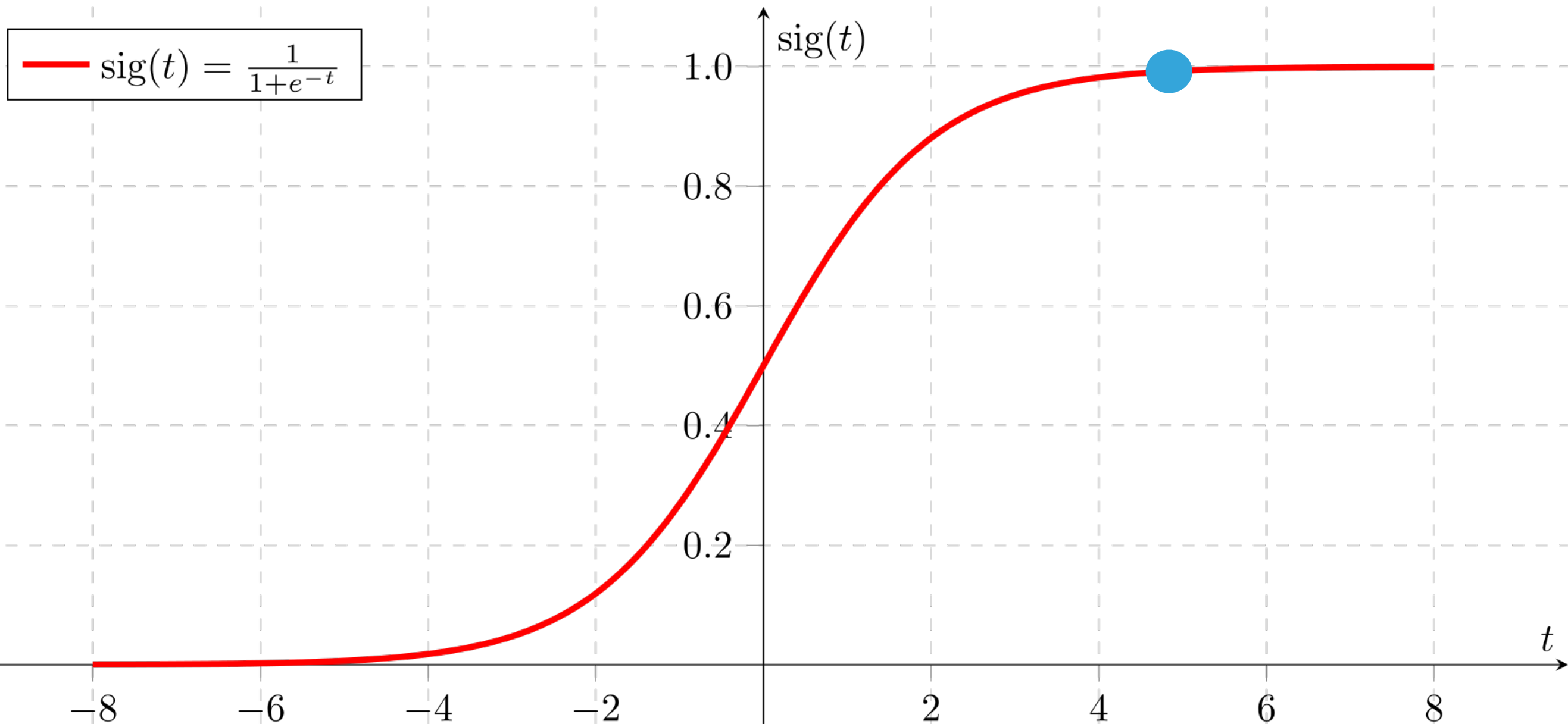
These values may be interpreted as probabilities





SIGMOID EQUATION

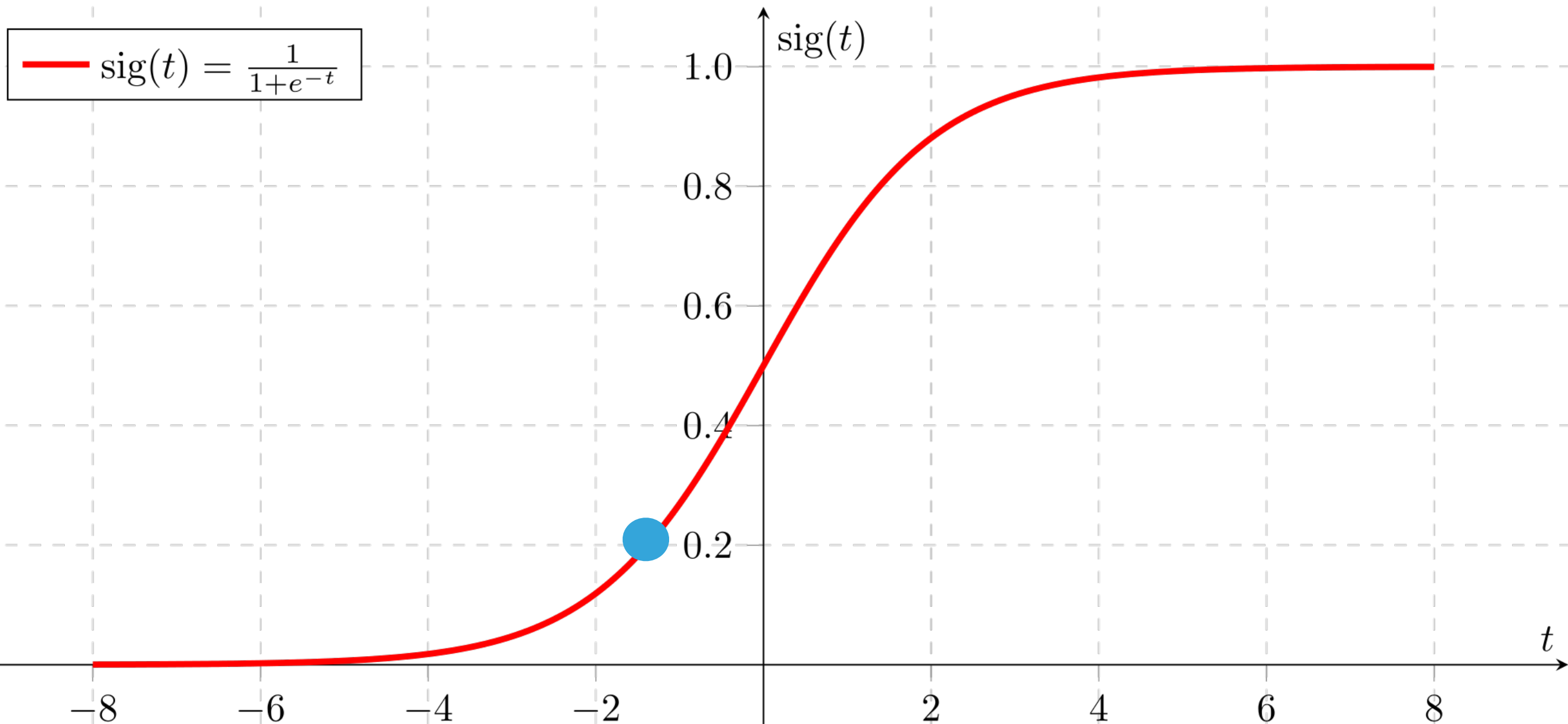
This point has a y-value of 0.98 so there is a 98% probability it belongs in class 1.





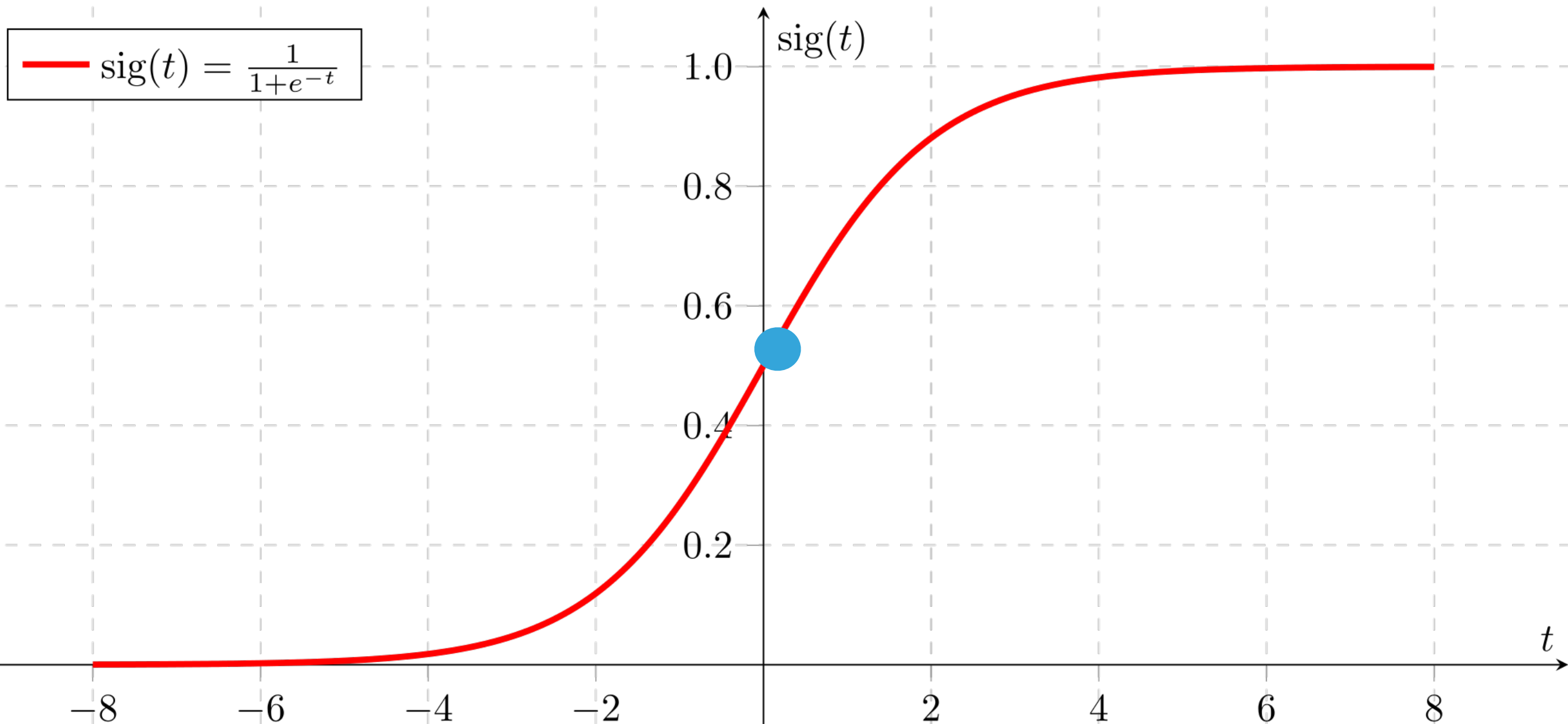
SIGMOID EQUATION

This point has a y-value of 0.2 so there is a 20% probability it belongs in class 1, and 80% probability it belongs to class 0.



SIGMOID EQUATION

This point is almost indeterminate, but it will be placed in class 1 since there is a probability of greater than 50% that it belongs in class 1.



HOW LOGISTIC REGRESSION WORKS

- ▶ Weights are randomly chosen to multiply each column.
- ▶ The weighted columns are summed.
- ▶ The sum is placed in the sigmoid equation and mapped to 0 or 1.
- ▶ Weights are adjusted depending on the error (using gradient descent).
- ▶ Another iteration continues.
- ▶ The iterations continue until the errors are minimized.

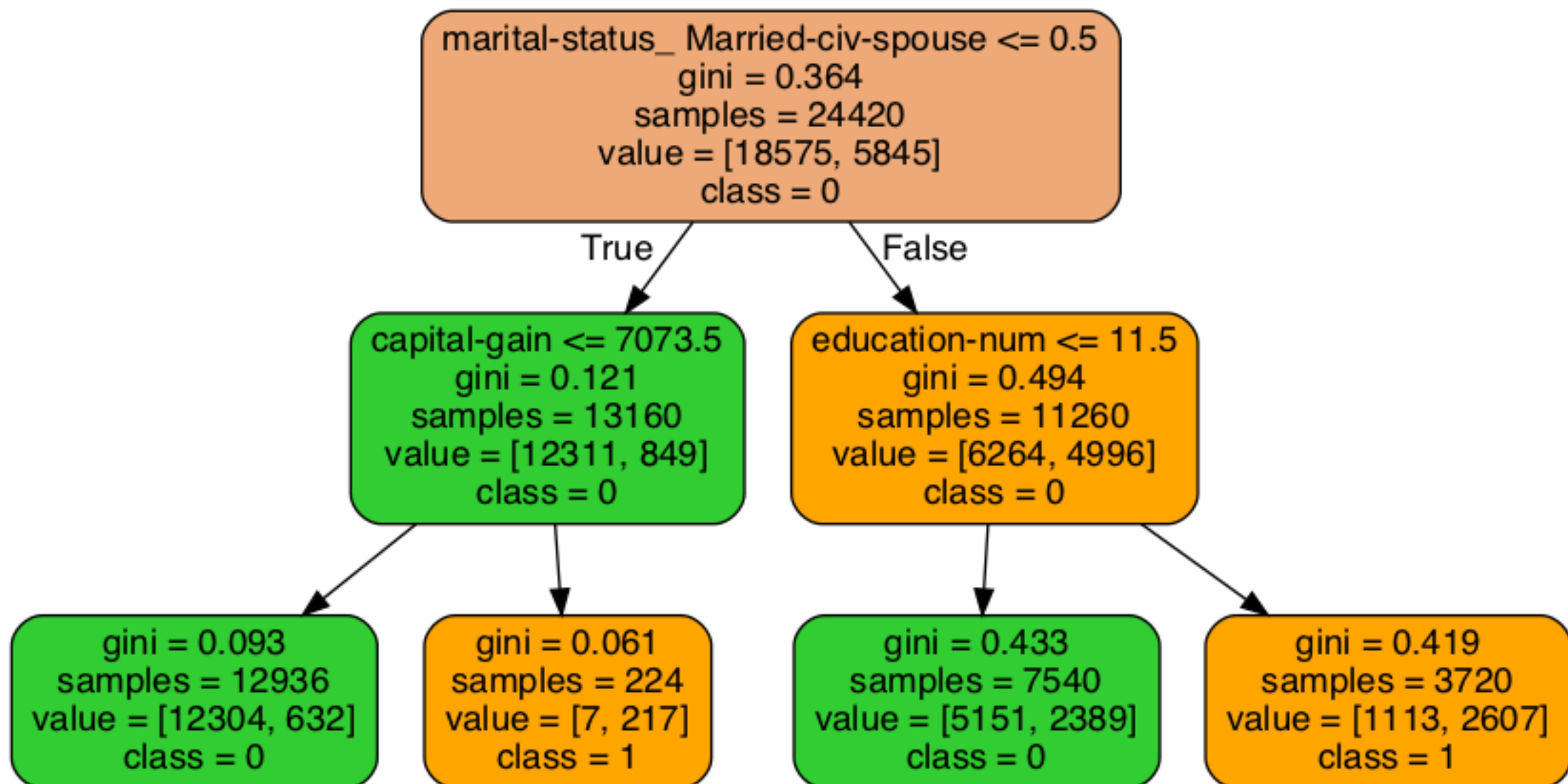
DECISION TREES

DECISION TREES

- ▶ Like a game of 20 Questions
- ▶ Each round, the data is split into two groups.
- ▶ Splits are made by determining which split will give the best score each round.
- ▶ The tree stops splitting when a desired level of accuracy has been reached.

DECISION TREE

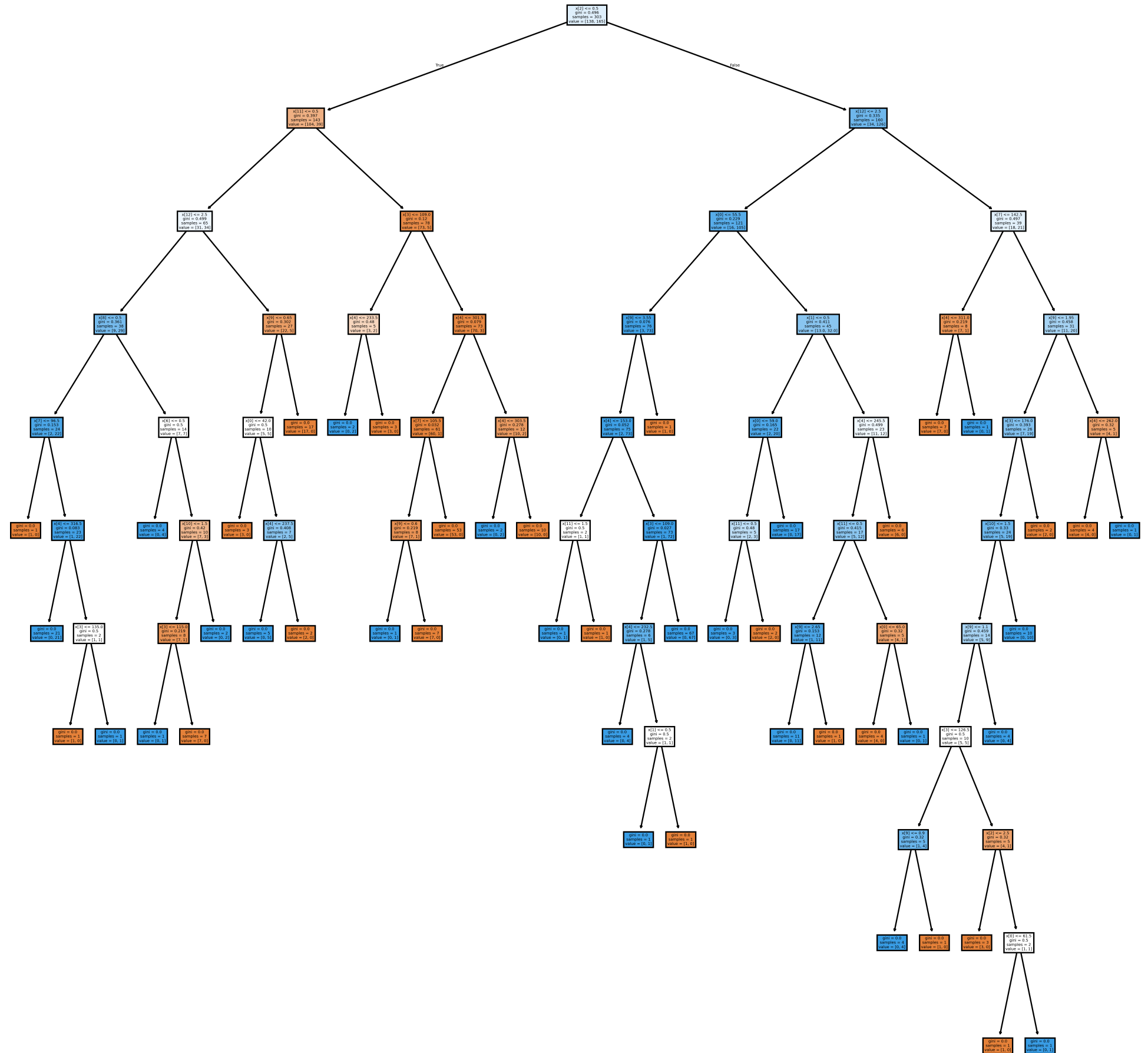
Census Dataset - max_depth=2





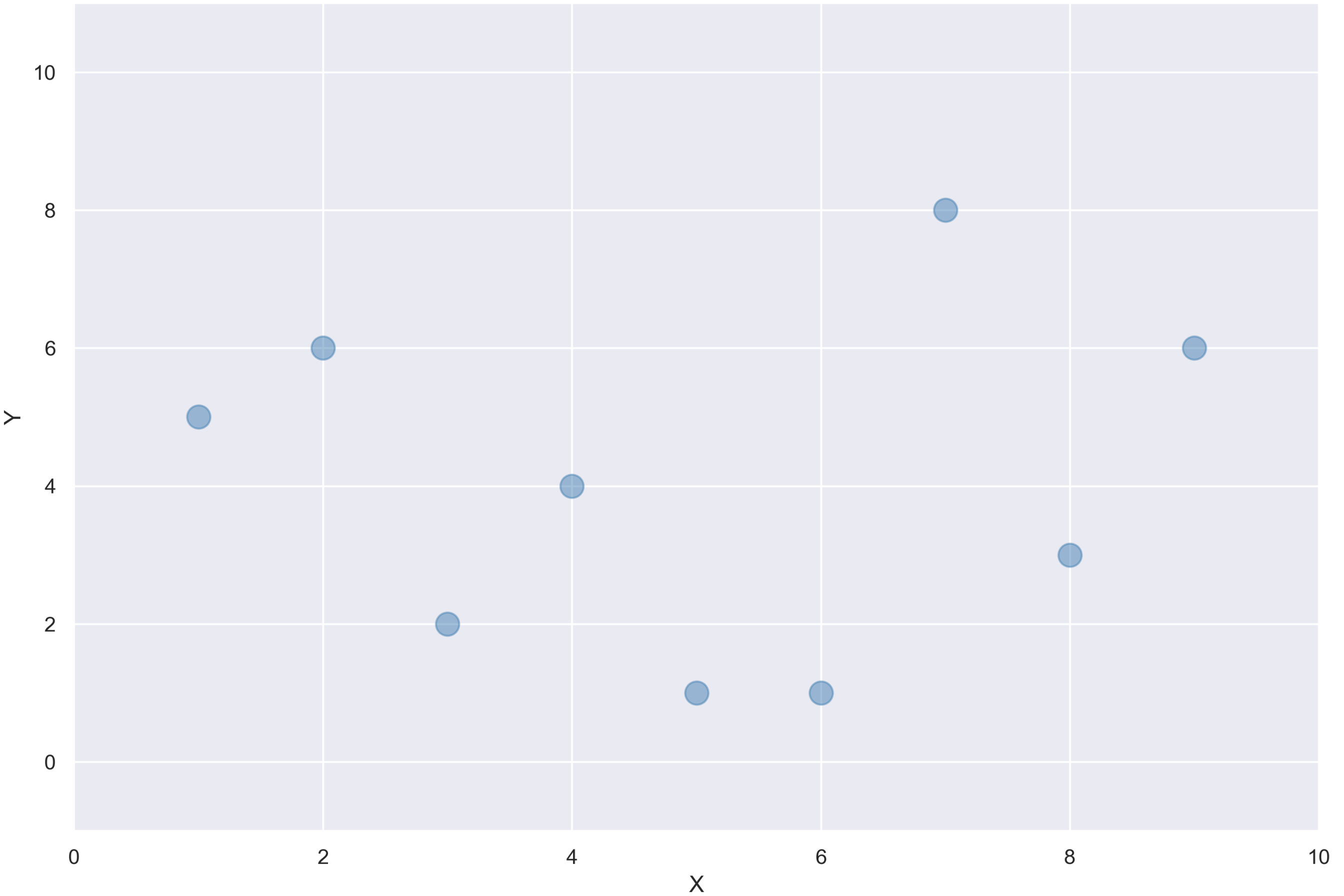
DECISION TREE

Heart Dataset
no max-depth

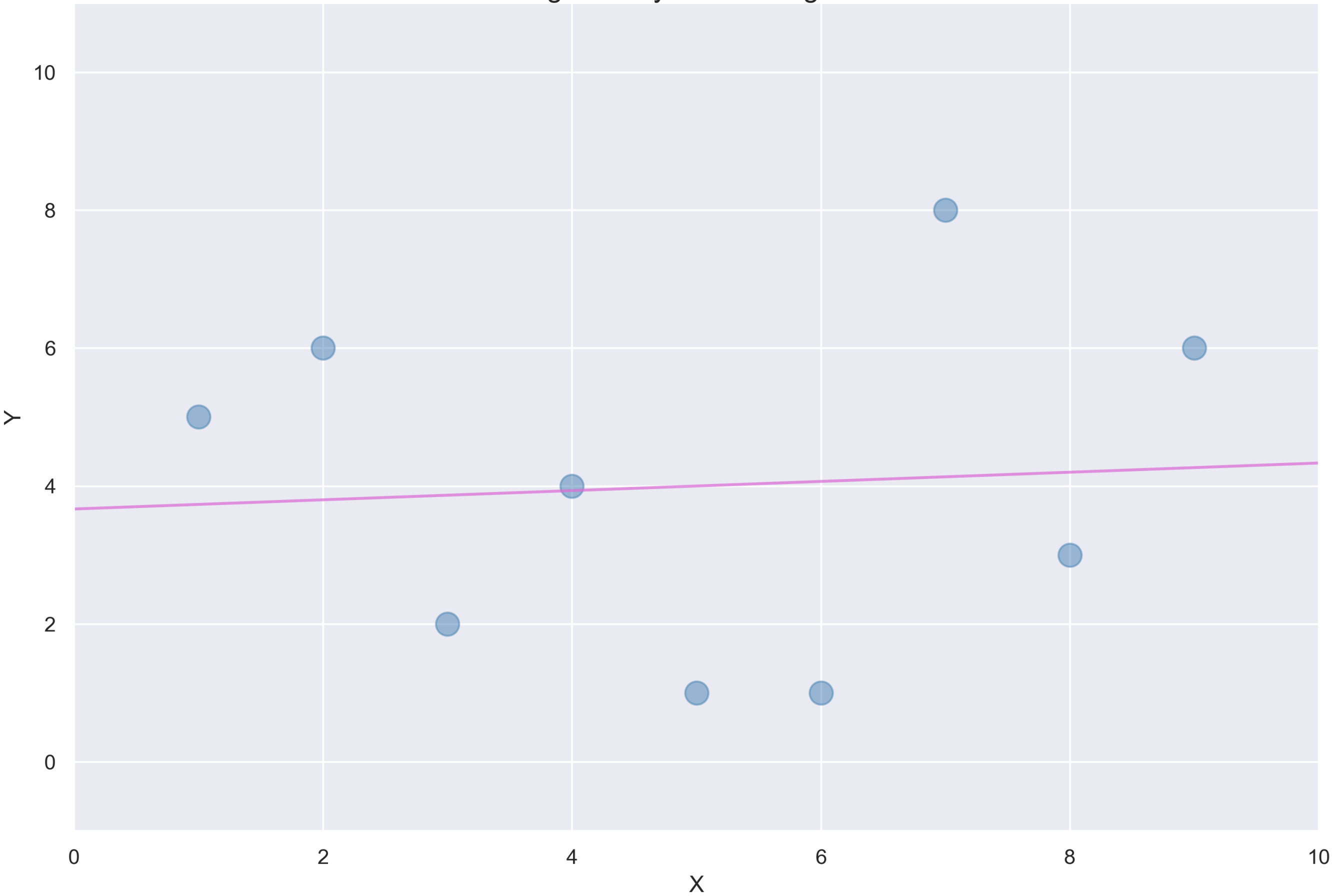


OVERFITTING THE DATA

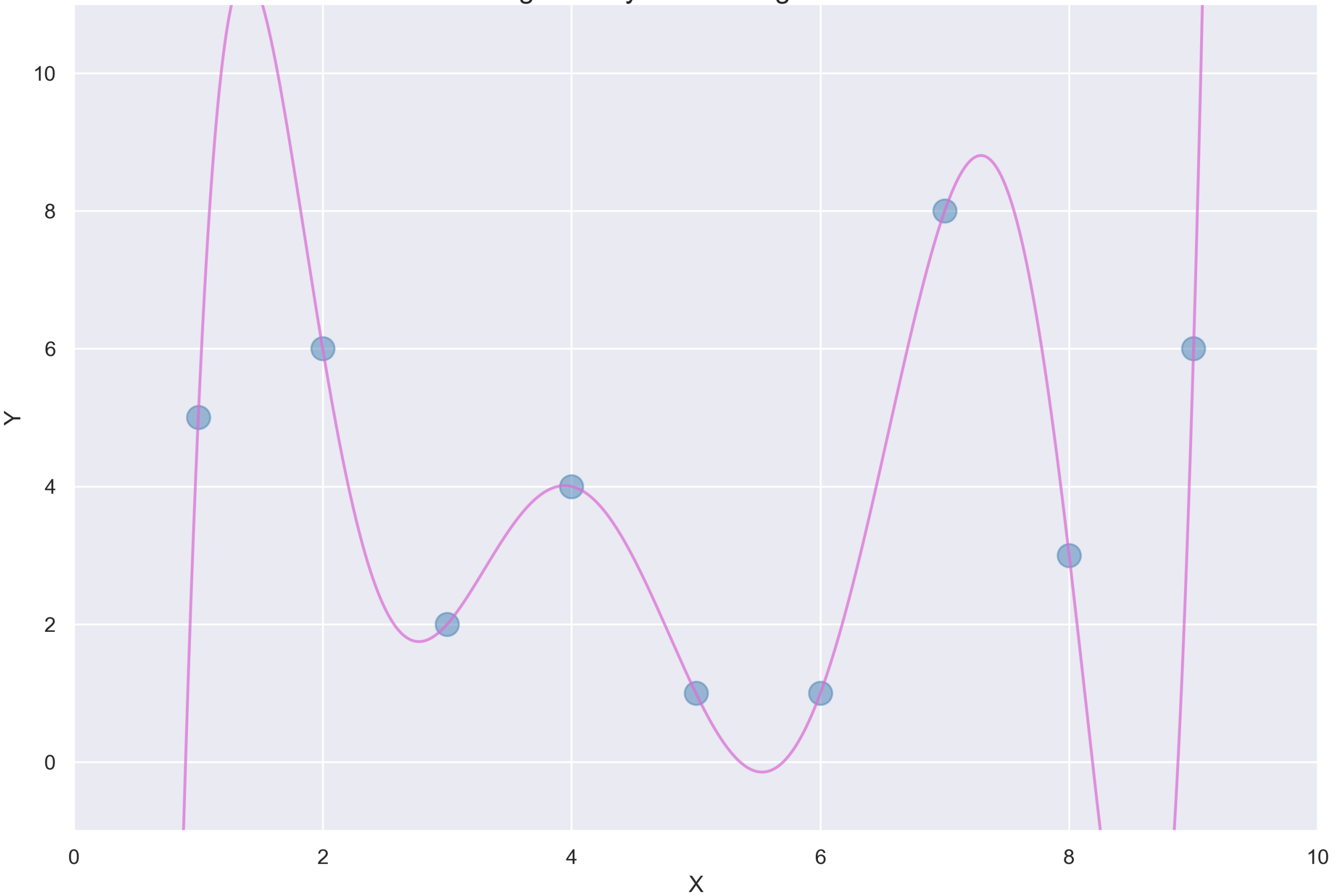
Random Points



1st Degree Polynomial - High Bias

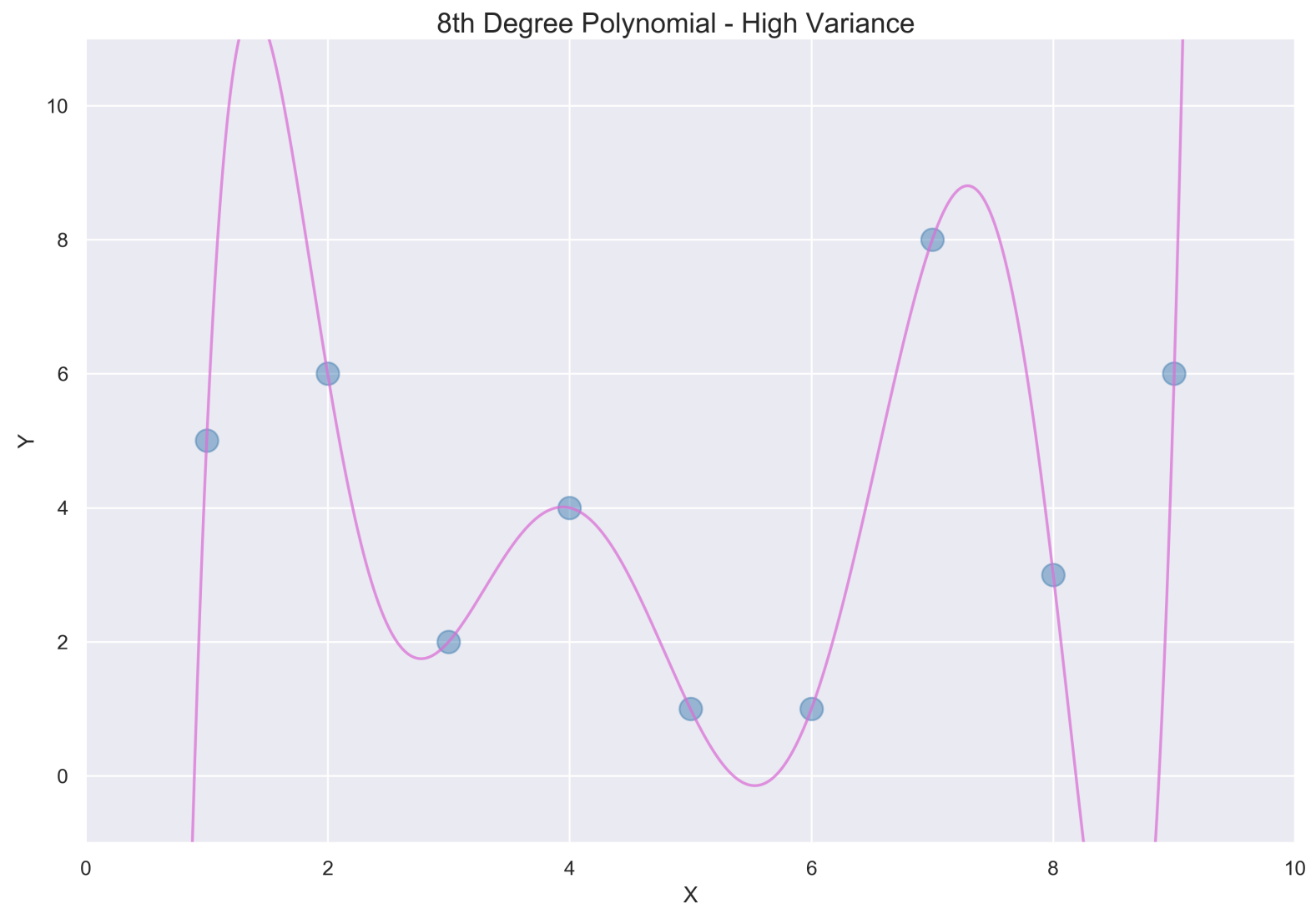


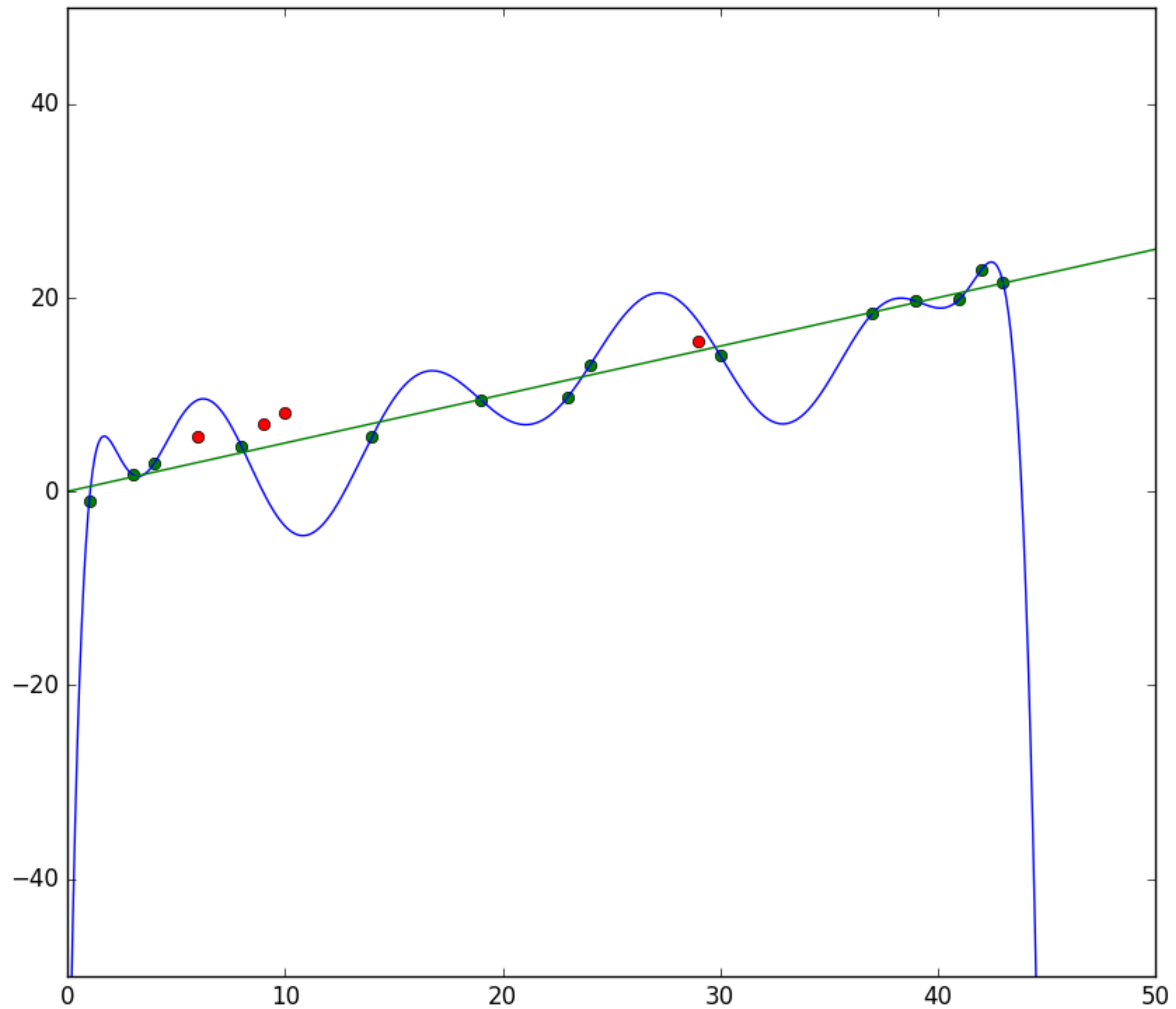
8th Degree Polynomial - High Variance

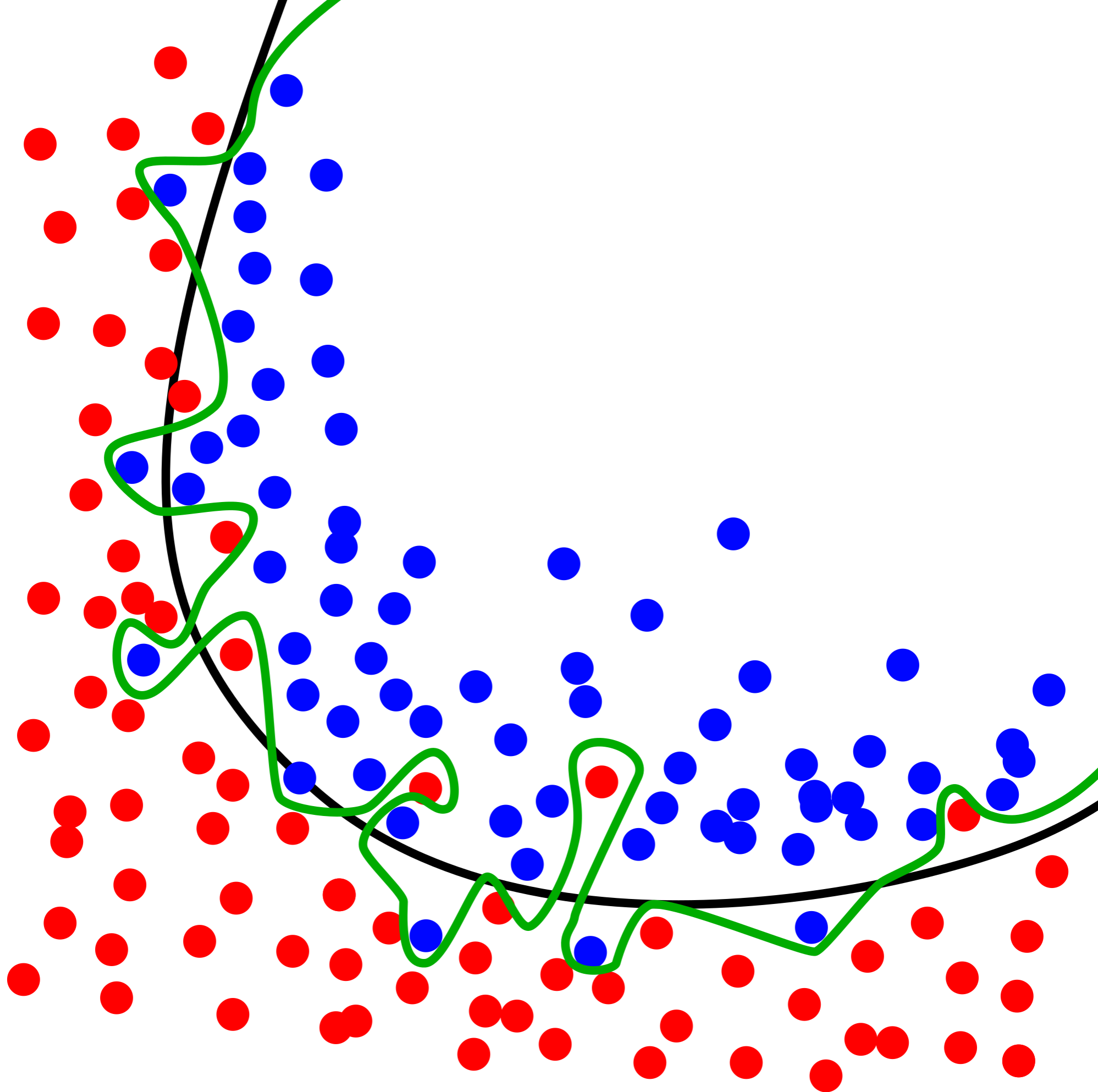


OVERFITTING

- ▶ Polynomial generalizes poorly to new data.
- ▶ When $x=1.5$, $y=11$.
- ▶ When $x=8.5$, $y=-3$.
- ▶ When $x=10$, $y=250$.

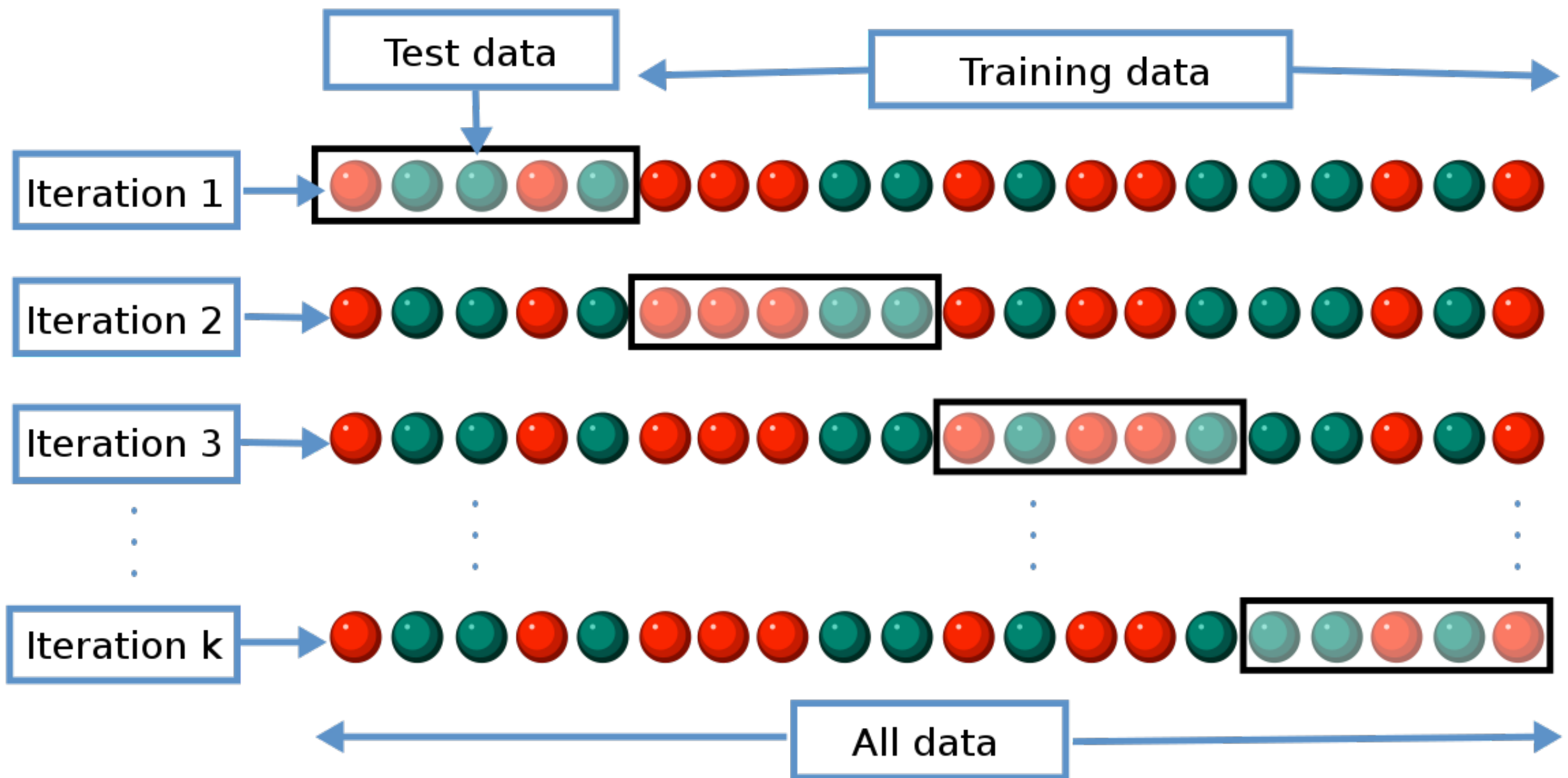






CROSS-VALIDATION

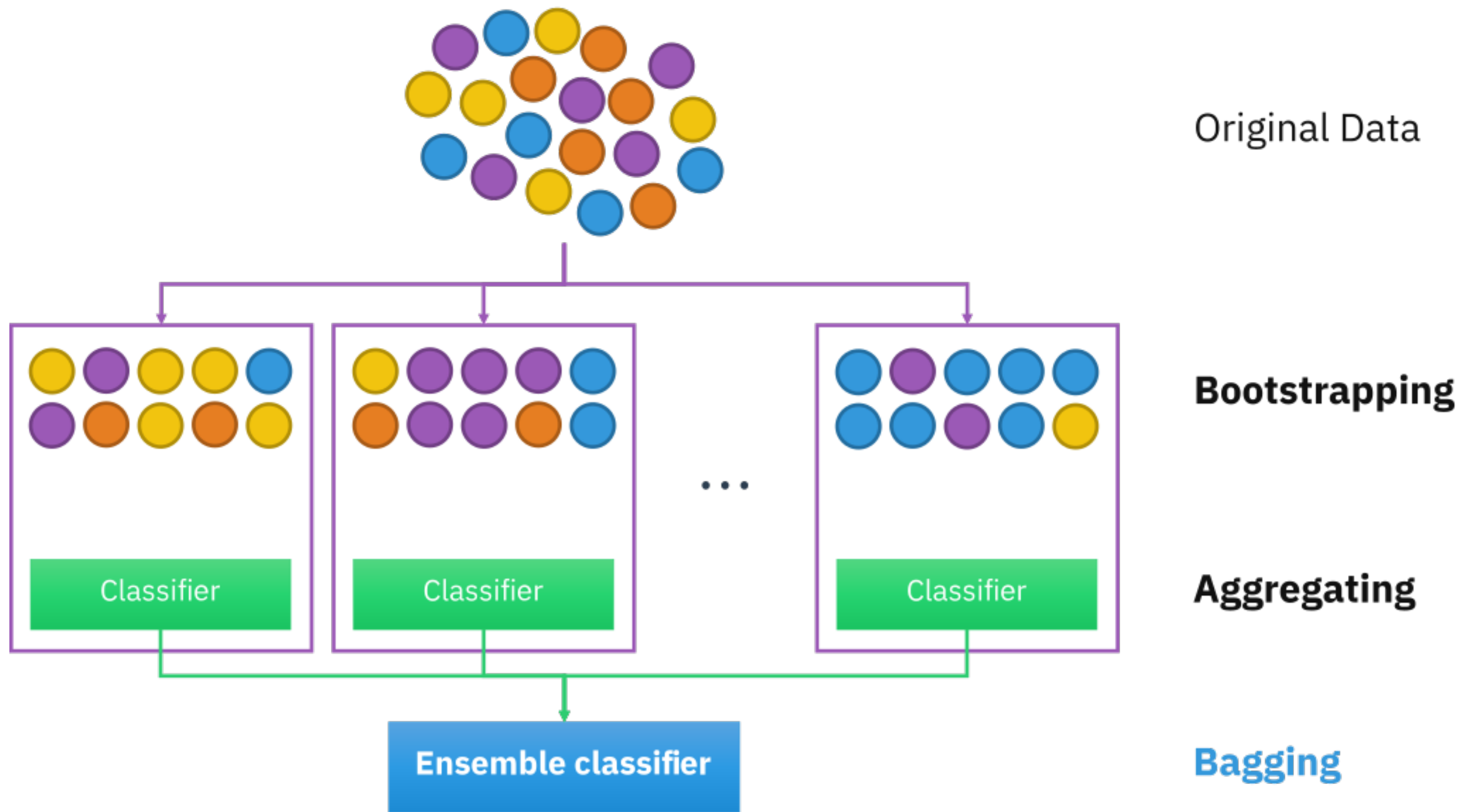
CROSS-VALIDATION IMAGE



RANDOM FORESTS

RANDOM FORESTS

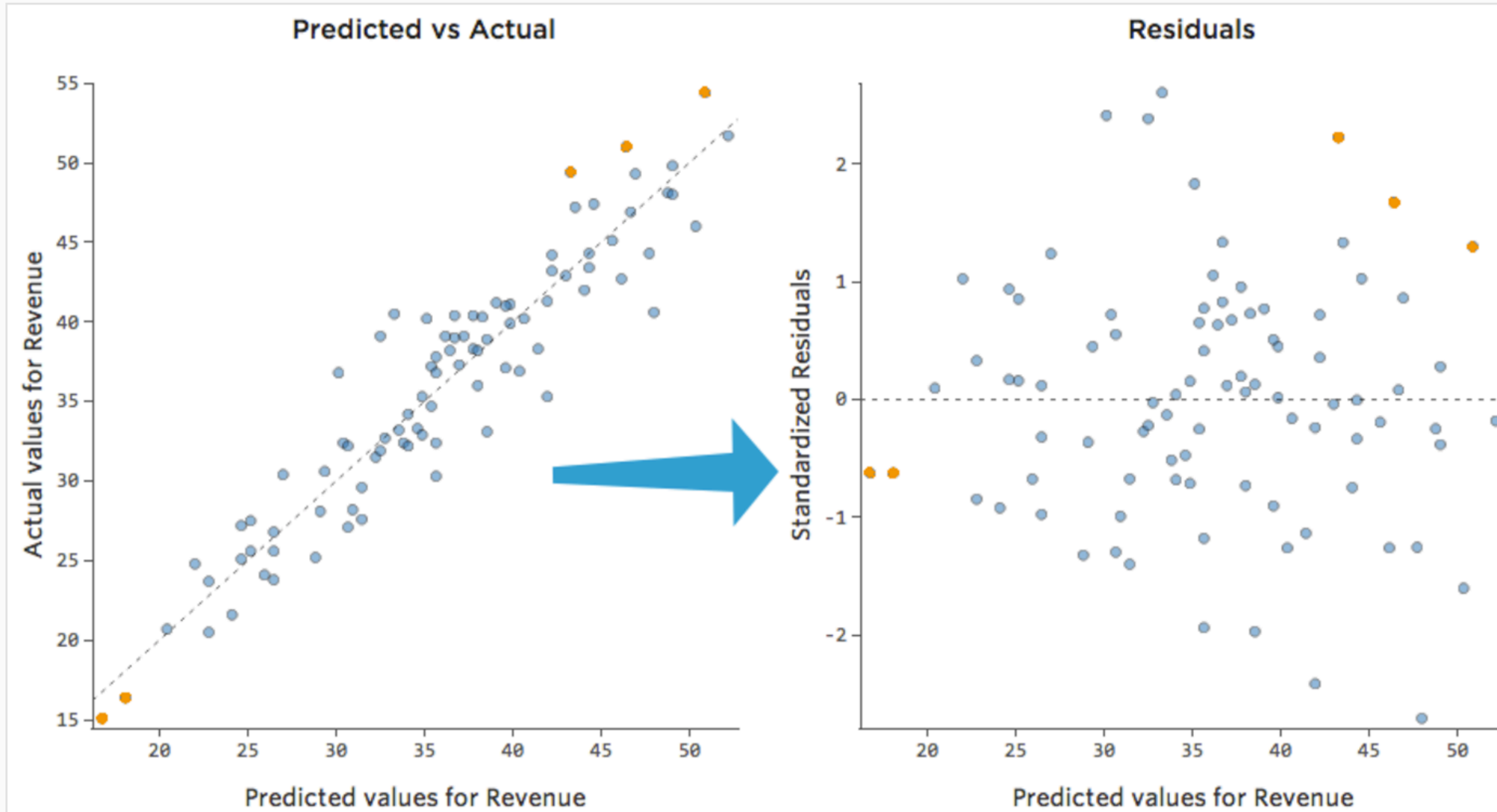
- ▶ Combine many Decision Trees (100 by Default).
- ▶ Trees are bootstrapped, meaning sampled with replacement.
- ▶ Randomness is involved with choosing the rows (bootstrapping) and the number of features (columns) to consider when making tree splits.
- ▶ Since the Decision Trees vary, they give different predictions.
- ▶ The Random Forest takes the average of all trees to make a final prediction.
- ▶ Random Forests use bagging: bootstrapping + aggregation.



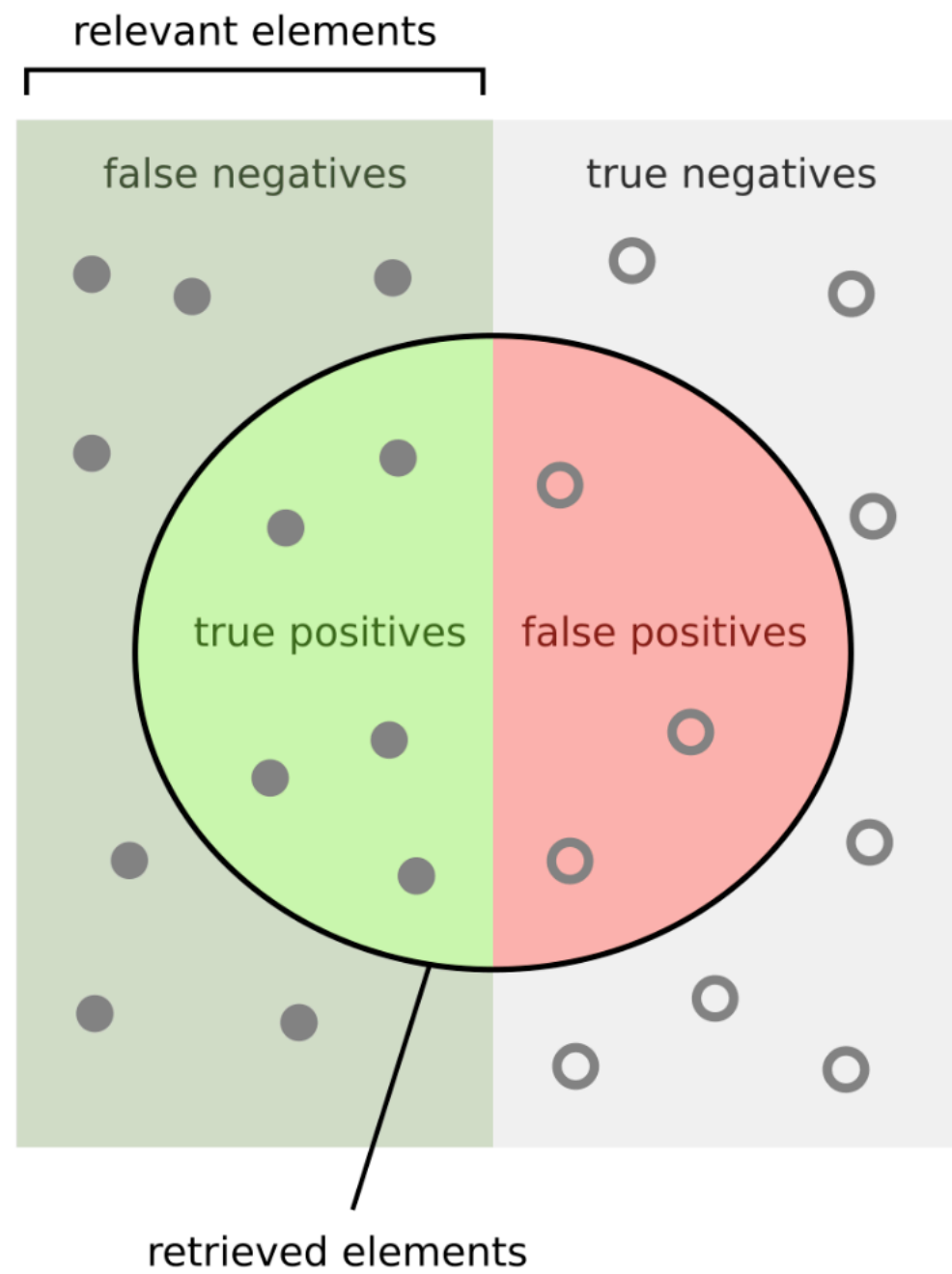
GRADIENT BOOSTING

GRADIENT BOOSTING

- ▶ Like Random Forests, Gradient Boosting is an ensemble of Decision Trees.
- ▶ Unlike Random Forests, Gradient Boosting trees evolve with each iteration.
- ▶ Each new tree is trained on the errors (the residuals) of the previous tree.
- ▶ By training on the errors, boosted trees can build upon small gains.
- ▶ Gradient Boosting can turn a weak learner into a strong learner.
- ▶ Boosting has been shown to be more effective than bagging overall.



Note that we've colored in a few dots in orange so you can get the sense of how this transformation works.

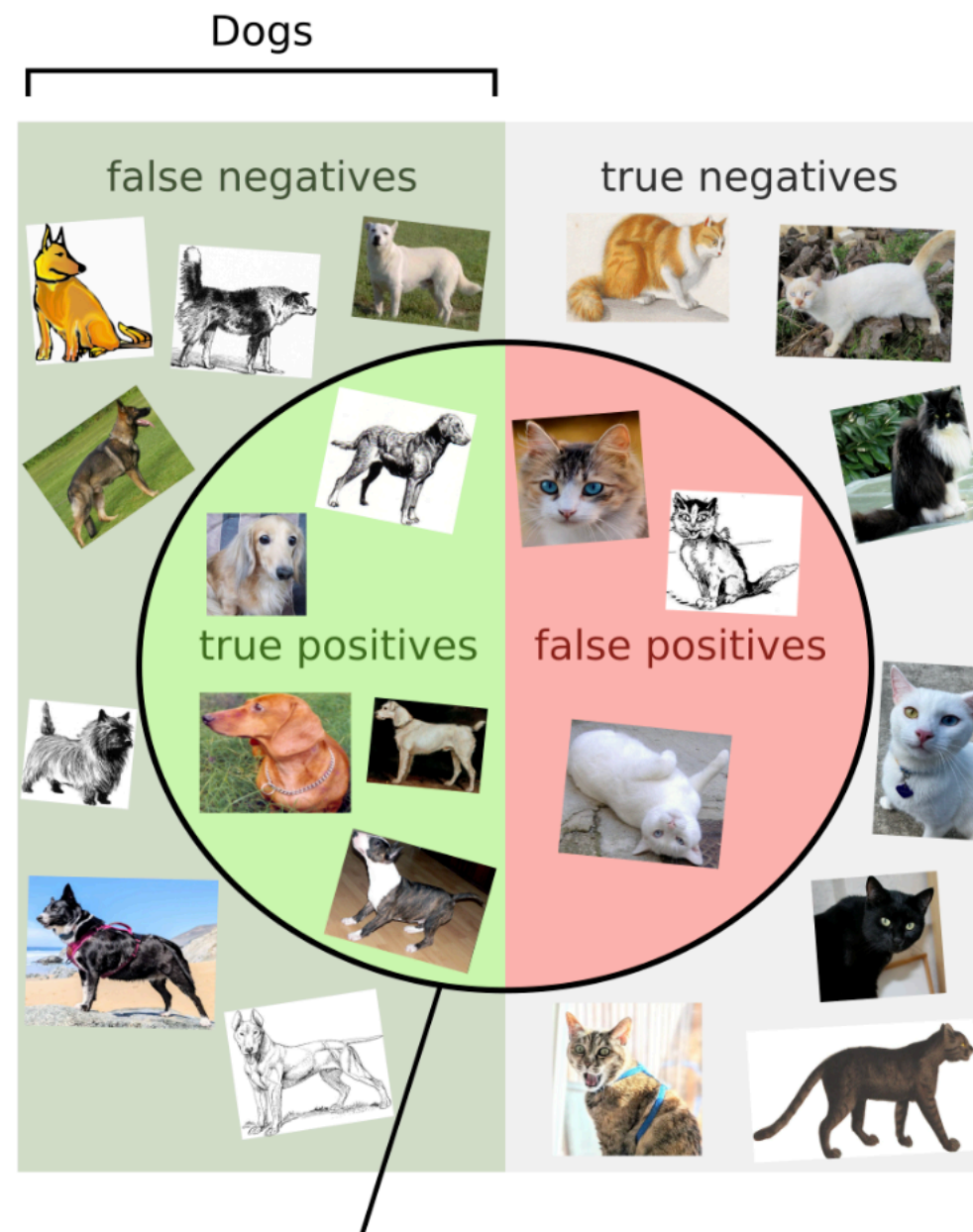


How many retrieved items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant items are retrieved?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$



$$\text{Precision} = \frac{5 \text{ true pos.}}{8 \text{ total pos.}} \quad \text{Recall} = \frac{5 \text{ true pos.}}{12 \text{ total dogs}}$$

$$\text{Prevalence} = \frac{12 \text{ total dogs}}{22 \text{ total images}}$$

$$\text{Accuracy} = \frac{5 \text{ true pos.} + 7 \text{ true neg.}}{22 \text{ total images}}$$



RECOMMENDED REFERENCES

- ▶ MIT *Machine Learning with Python* online course starting May 27 - free without certificate: <https://learn.mit.edu/search?free=true&resource=3155>
- ▶ Deep Learning MIT 2026 online lectures: https://www.youtube.com/playlist?list=PLtBw6njQRU-rwp5__7C0oIVt26ZgjG9NI
- ▶ O'Reilly Live Courses: <https://www.oreilly.com/live/>
- ▶ Jason Brownlee: <https://machinelearningmastery.com/>
- ▶ BCA Playlists: <https://www.youtube.com/@berkeleycodingacademy7015/playlists>
- ▶ *Hands-on Gradient Boosting with XGBoost and Scikit-learn*
- ▶ *Hands-on Machine Learning with Scikit-Learn, Keras, and PyTorch*